Xiangyu (Hsiangyu) Zhao

PhD Student at Monash University

Multimodal LLMs for Mental Health | Medical Image Analysis | Translational AI for Clinical Practice

Monash Email: xiangyu.zhao@monash.edu | Personal Email: hsiangyu.zhao@outlook.com

Homepage | Google Scholar | LinkedIn | GitHub

EDUCATION

PhD, Data Science and Artificial Intelligence (in progress) — **2026 - Present**

Monash University | Melbourne, Australia

Master of Engineering, Electronic and Information Engineering — **2021 - 2024**

Shanghai Jiao Tong University | Shanghai, China

GPA: 3.78/4.00 | Shanghai Outstanding Graduate

Bachelor of Engineering, Biomedical Engineering — **2017 - 2021**

Beihang University | Beijing, China

GPA: 3.87/4.00 | Rank: 1/64 | Beijing Outstanding Graduate

RESEARCH EXPERIENCE

Research Intern — **Aug 2025 - Feb 2026**

Xiaohongshu (RED) | Shanghai, China

- Worked on multimodal LLM post-training.

Research Assistant — **Jun 2024 - Jan 2026**

Monash Suzhou | Suzhou, China / Hybrid

- Developed multimodal and language-model research for mental health, including audio-visual depression detection, safety and ethics evaluation, audio-grounded process evaluation, and clinician-oriented explanations.
- Collaborated across machine learning, psychiatry, psychology, and clinical research teams.

Research Intern — **Aug 2023 - Jun 2024**

United Imaging Intelligence | Shanghai, China

- Developed foundation-model methods for medical image segmentation, including general multimodal segmentation and learning from partially labeled datasets.

Research Intern — **Feb 2021 - Jun 2021**

Institute of Automation, Chinese Academy of Sciences | Beijing, China

- Studied object detection for large-scale images.

SELECTED PUBLICATIONS

Selected first-author peer-reviewed and public research outputs. Full list: Google Scholar.

1. **Xiangyu Zhao**, Junyu Yan, Yaling Shen, Stephanie Fong, Zimu Wang, Yiwen Jiang, Qingyang Xu, Jiahe Liu, Dominic Dwyer, and Zongyuan Ge. "AudioProcessBench: Benchmark for Identifying Process Errors in Audio-Grounded Reasoning." *Findings of EMNLP (accepted)*, 2026.
2. **Xiangyu Zhao**, Yaling Shen, Yiwen Jiang, Zimu Wang, Jiahe Liu, Maxmartwell H. Cheng, Guilherme C. Oliveira, Robert Desimone, Dominic Dwyer, and Zongyuan Ge. "It Hears, It Sees Too: Multi-Modal LLM for Depression Detection by Integrating Visual Understanding into Audio Language Models." *Findings of AACL-IJCNLP (accepted)*, 2026.
3. **Xiangyu Zhao**, Sheng Wang, Zhiyun Song, Zhenrong Shen, Linlin Yao, Haolei Yuan, Qian Wang, and Lichi Zhang. "AdLER: Adversarial Training with Label Error Rectification for One-Shot Medical Image Segmentation." *Expert Systems with Applications, 310:131224*, 2026.
4. **Xiangyu Zhao**, Xi Ouyang, Lichi Zhang, Zhong Xue, and Dinggang Shen. "Exploiting Latent Classes for Medical Image Segmentation from Partially Labeled Datasets." *MICCAI*, 2024.
5. **Xiangyu Zhao**, Di Zang, Sheng Wang, Zhenrong Shen, Kai Xuan, Zeyu Wei, Zhe Wang, Ruizhe Zheng, Xuehai Wu, Zheren Li, Qian Wang, Zengxin Qi, and Lichi Zhang. "sTBI-GAN: An Adversarial Learning Approach for Data Synthesis on Traumatic Brain Segmentation." *Computerized Medical Imaging and Graphics, 112:102325*, 2024.
6. **Xiangyu Zhao**, Zengxin Qi, Sheng Wang, Qian Wang, Xuehai Wu, Ying Mao, and Lichi Zhang. "RCPS: Rectified Contrastive Pseudo Supervision for Semi-Supervised Medical Image Segmentation." *IEEE Journal of Biomedical and Health Informatics, 28(1):251-261*, 2023.

7. **Xiangyu Zhao**, Zhenrong Shen, Dongdong Chen, Sheng Wang, Zixu Zhuang, Qian Wang, and Lichi Zhang. "One-Shot Traumatic Brain Segmentation with Adversarial Training and Uncertainty Rectification." *MICCAI*, 2023.

8. **Xiangyu Zhao**, Peng Zhang, Fan Song, Chenbin Ma, Guangda Fan, Yangyang Sun, Youdan Feng, and Guanglei Zhang. "Prior Attention Network for Multi-Lesion Segmentation in Medical Images." *IEEE Transactions on Medical Imaging, 41(12):3812-3823*, 2022.

9. **Xiangyu Zhao**, Peng Zhang, Fan Song, Guangda Fan, Yangyang Sun, Yujia Wang, Zheyuan Tian, Luqi Zhang, and Guanglei Zhang. "D2A U-Net: Automatic Segmentation of COVID-19 CT Slices Based on Dual Attention and Hybrid Dilated Convolution." *Computers in Biology and Medicine, 135:104526*, 2021.

HONORS AND AWARDS

Shanghai Outstanding Graduate **2024**

First Prize, Master Student Scholarship, Shanghai Jiao Tong University **2022**

Beijing Outstanding Graduate **2021**

Meritorious Winner, COMAP Mathematical Contest in Modeling **2020**

Second Prize (China) / First Prize (Beijing), CUMCM **2019**

Merit Student and Academic Excellence Scholarship, Beihang University (three times) **2017 - 2020**

TEACHING AND SERVICE

Teaching Assistant, Computer Vision in Biomedical Engineering **Spring 2022**
Shanghai Jiao Tong University | Shanghai, China

Student Mentor **2018 - 2019**
Beihang University | Beijing, China